Course Outline

- Abstract Data Types (ADTs):
  - Lists, Stacks, Queues;
  - Trees: Binary, binary search, tree traversal, AVL, Multi-way trees, B Trees, heaps;
  - Graphs: depth-first-search, breadth-first-search, shortest paths, spanning tree;
- Algorithms: Sorting and searching, Traversals, Dictionary, Topological ordering, Graphs, and different Problem solving approaches.

- Textbook: Data Structures and Algorithms in Python -- Michael T. Goodrich, Roberto Tamassia, and Michael H. Goldwasser, Wiley.

Grading Policy

First In-semester exam – 15%
Second In-semester exam – 15%
End Semester Exam – 30%
Project, presentation – 20%
Lab practice and performance – 20%

Why Data Structures ?

Real-world Practice

Software applications/Developments

- Problem definition, specification, requirement
- Solution approaches (**design**)
- Implementation of the solution (**coding**)
- Analysis of the solution (**measure performance**)
- Testing, debugging, and maintenance

Specification of a problem

A precise statement/description of the problem.

- It involves describing the input, the expected output, and the logic to get the output from the input.

Solution approaches

- Formulation of a method
  - a sequence of steps to solve the problem (algorithm).
- The design can be done by pseudo-code, flowcharts, steps, or any combinations of these.
- During the design phase the problem can be decomposed into smaller pieces, where each piece is designed into a *module*. → Divide and Conquer

Implementation of the design

- Development of code using a programming language will make the design functions testable.

- The design and implementation of the problem require **data structures** that uses data, operations and how they are being implemented to perform operations on data.

Implementation of the design

- Development of code using a programming language will make the design functions efficiently.

- The design and implementation of the problem require **data structures** that uses data, operations and how they are being implemented to perform operations on data.

How much time and memory an algorithm consumes while experimenting the design and finding intended outcomes ?

What is Data Structures ?